

الرسم بالحاسب

3:5

الثلاثاء 30/3/2021

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Faculty of Computers & Information, Assiut University

3rd Level

Final Exam

Duration: 2 hours

* Required

* This form will record your name, please fill your name.

1

* الإسم الرباعي (بالعربي فقط)

2

* رقم الجلوس

3

* المستوى

☐ الاول

☐ الثاني

☐ الثالث

☐ الرابع

4

* البرنامج

☐ عام

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☐ هندسة

* رقم المعمل

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6

* رقم الكمبيوتر

7

* كود المراقب

8

* إسم الملف بالمسار (تملي بمعرفة المراقب)

9

..... light is so scattered; it is impossible to tell what its original source was *
(1 Point)

- ☐ Diffuse
- ☐ Specular
- ☒ Ambient
- ☐ Spot

10

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);
```

```
glLoadIdentity();
```

```
glRotatef(90.0, 0, 0, 1.0);
```

```
glRotatef(90.0, 0, 0, 1.0); *
```

(1 Point)

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- ☐ A
- ☐ B
- ☒ C
- ☐ D

11

..... projection preserves parallel lines. *
(1 Point)

- ☒ Orthographic
- ☐ Perspective
- ☐ Frustum
- ☐ Symmetric

12

What are the four major types of shaders available in OpenGL 4.0? *
(1 Point)

- ☐ Filtered, Non-filtered, Exponent and Fragmented
- ☐ Multi-sampled, Single-sampled, Over-sampled and 16x
- ☐ Vertex, Geometry, Isomorphic and Anisotropic
- ☒ Geometry, Tessellation, Vertex and Fragment

13

Once a 2D or 3D rotation is performed, is it possible to reverse the rotation? *
(1 Point)

- ☒ Yes
- ☐ No

14

The volume of a small cube change when its vertices are rotated *
(1 Point)

☐ True

☒ False

15

What is the dimension of a 3D transformation matrix in OpenGL? *
(1 Point)

☐ 2×2

☐ 3×3

☐ 3×4

☒ 4×4

16

If the GL_SPOT_CUTOFF is 180, spotlight effect equals *
(1 Point)

☐ 0

☒ 1

☐ 2

☐ 3

17

With color tracking, you can tell OpenGL to set material properties by only calling glColor. *

(1 Point)

☒ True

☐ False

18

Which function call sets up the size of the output area? *

(1 Point)

☒ glViewport()

☐ gluPerspective()

☐ glDisplayfunc()

☐ None of These

19

Is it possible to perform a reflection of an object across a specified axis? *

(1 Point)

☐ No, this functionality is not possible with the current version of OpenGL

☐ Yes, with glReflect()

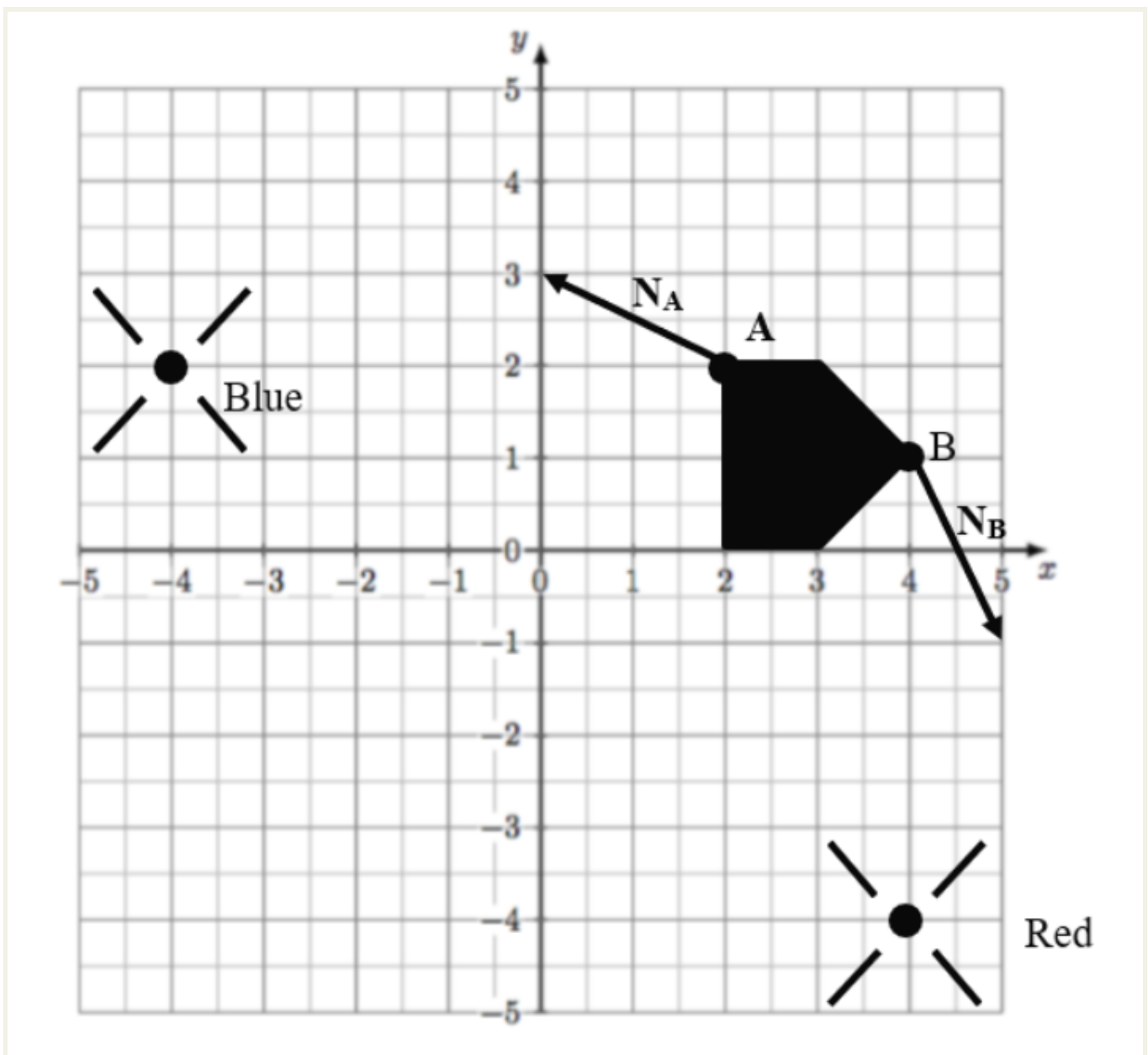
☐ No, you must redraw your object reflected across the desired axis

☒ Yes with simple matrix multiplication

The figure below contains two light sources and two points in 2D. The red-light source emits diffuse light = (1.0, 0.2, 0.2) and its spotlight effect=1. The blue-light source emits diffuse light = (0.2, 0.1, 1.0), spot cutoff = 45, and spot exponent=2.0. The normal vectors are shown in the figure. Assume the constant attenuation=0, linear attenuation =1.0, quadratic attenuation=0, the material ambient = (0, 0, 0), the material emission= (0.1,0.1,0.1), the diffuse components of the material = (0.5,1.0,0.5), and the specular components of the material = (0, 0, 0) for the two points. Using the Phong reflection model, answer the following questions:

The normalized normal vector $N_A = *$

(1 Point)



☐ $\left(\frac{-1}{\sqrt{3}}, \frac{2}{\sqrt{3}}\right)$

☐ $\left(\frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right)$

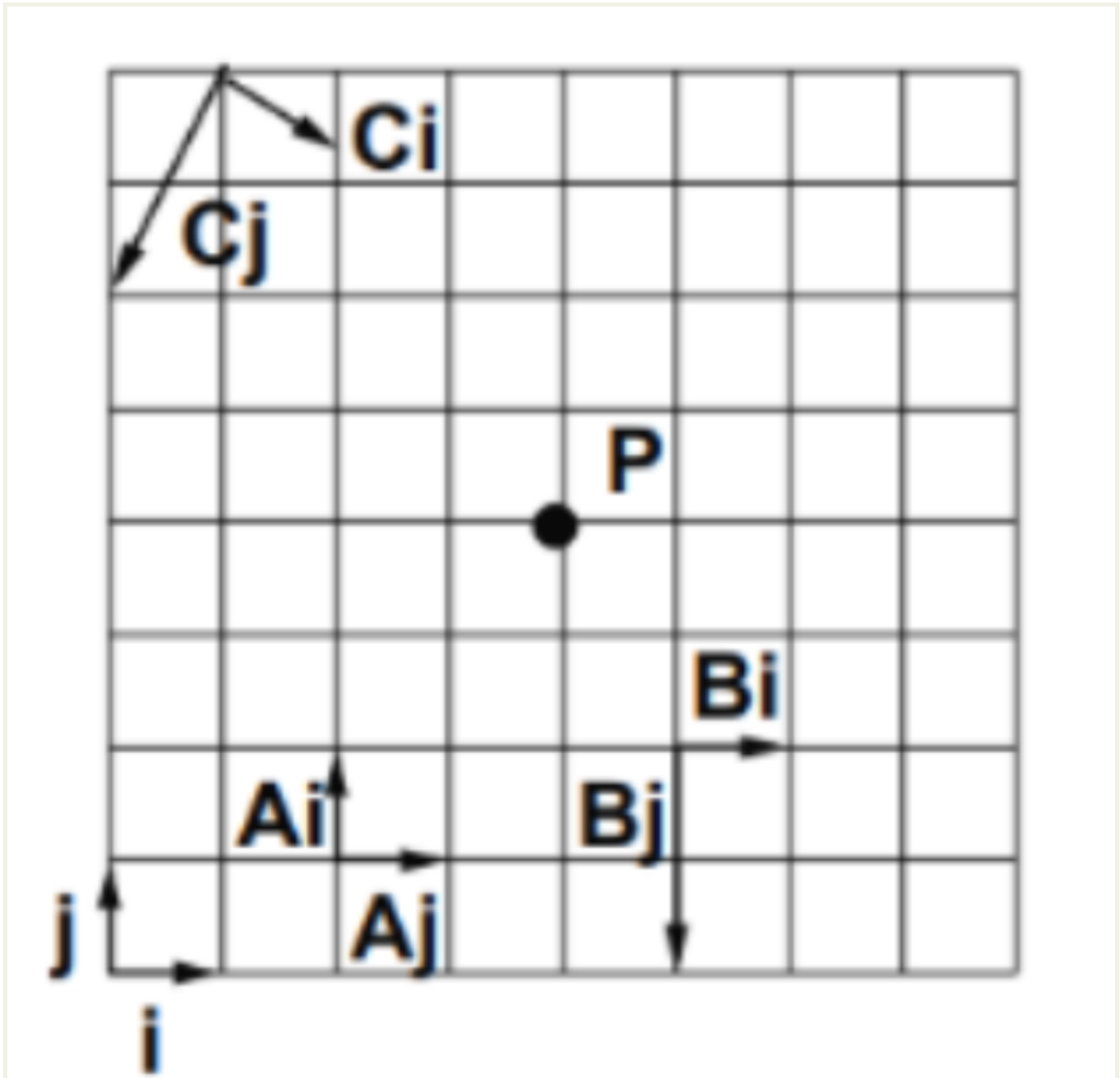
☐ $\left(\frac{-2}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$

☐ $\left(\frac{-2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right)$

The figure contains a point coordinate P that can be expressed as $P = 4*i + 4*j$, where i and j are basis vectors of unit length along the x and y axes, respectively. Using the figure below to answer the following questions.

..... is the coordinates of point P with respect to coordinate frames A *

(1 Point)



☐ (2,3)

☐ (4,4)

☒ (3,2)

☐ (-2,3)

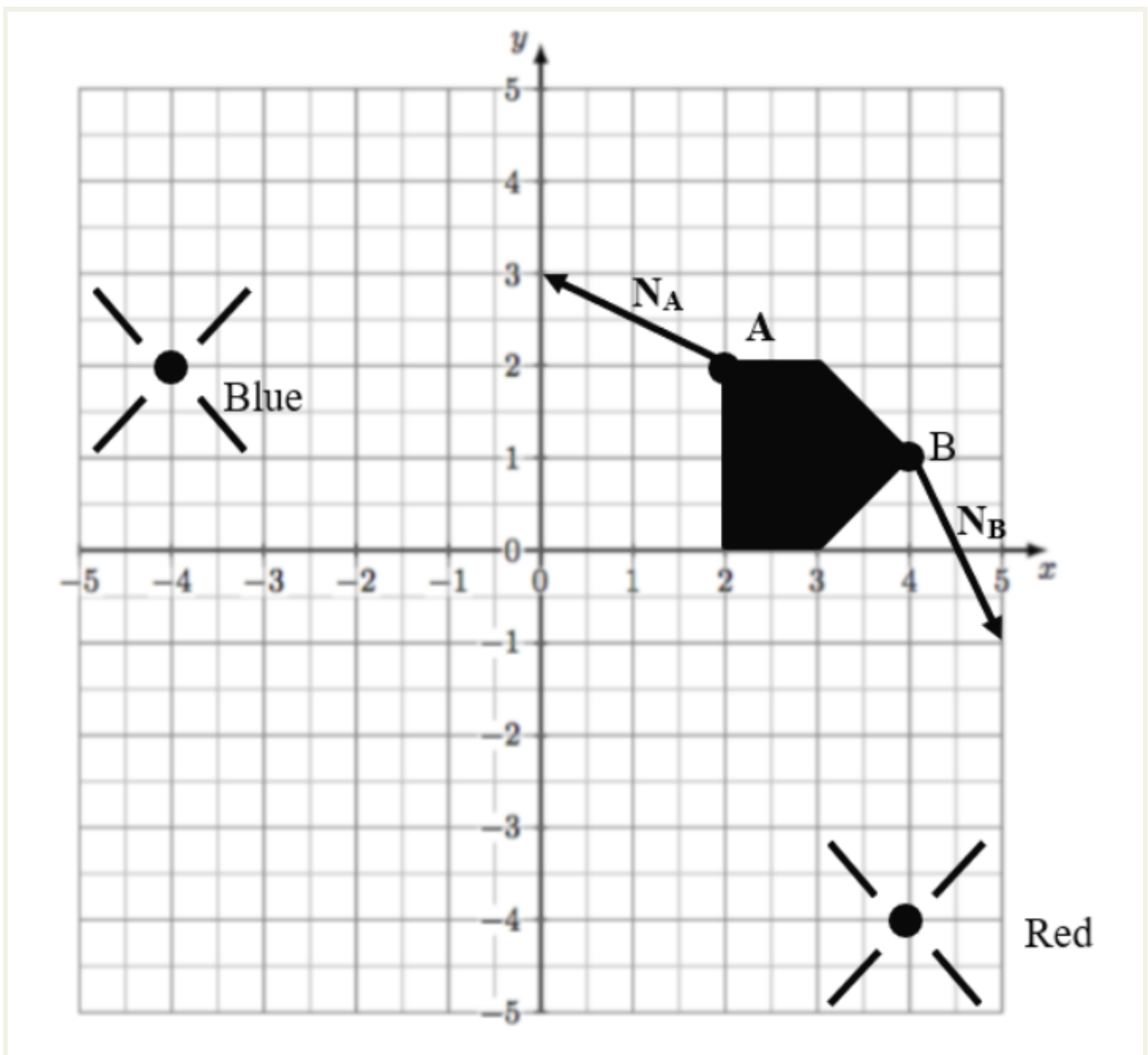
Double buffering solves which problem: *
(1 Point)

- ☐ producing stereo images
- ☐ remote debugging PCs
- ☒ displaying a frame buffer while it is being updated.
- ☐ Enhancing performance

The figure below contains two light sources and two points in 2D. The red-light source emits diffuse light = $(1.0, 0.2, 0.2)$ and its spotlight effect=1. The blue-light source emits diffuse light = $(0.2, 0.1, 1.0)$, spot cutoff = 45, and spot exponent=2.0. The normal vectors are shown in the figure. Assume the constant attenuation=0, linear attenuation =1.0, quadratic attenuation=0, the material ambient = $(0, 0, 0)$, the material emission= $(0.1, 0.1, 0.1)$, the diffuse components of the material = $(0.5, 1.0, 0.5)$, and the specular components of the material = $(0, 0, 0)$ for the two points. Using the Phong reflection model, answer the following questions:

The diffuse intensity for the point B for the red-light source *

(1 Point)



☐ $\frac{1}{2\sqrt{10}}$

☐ $\frac{3}{\sqrt{5}}$

☐ $\frac{2}{\sqrt{5}}$

☐ $\frac{3}{\sqrt{10}}$

24

What is the correct order of operations of the OpenGL graphics pipeline? *
(1 Point)

☐ projection transformation, modelview transformation, divide by w, viewport transform

☐ modelview transformation, divide by w, projection transformation, viewport transform

☐ modelview transformation, viewport transform, divide by w, projection transformation

☒ modelview transformation, projection transformation, divide by w, viewport transform

25

Given $U = (2, 1, -1)$ and $V = (2, -1, 0)$, the cross-product $U \times V$ equals *
(1 Point)

☐ (4, -1, 0)

☐ (1, 2, 0)

☐ (-4, 1, 0)

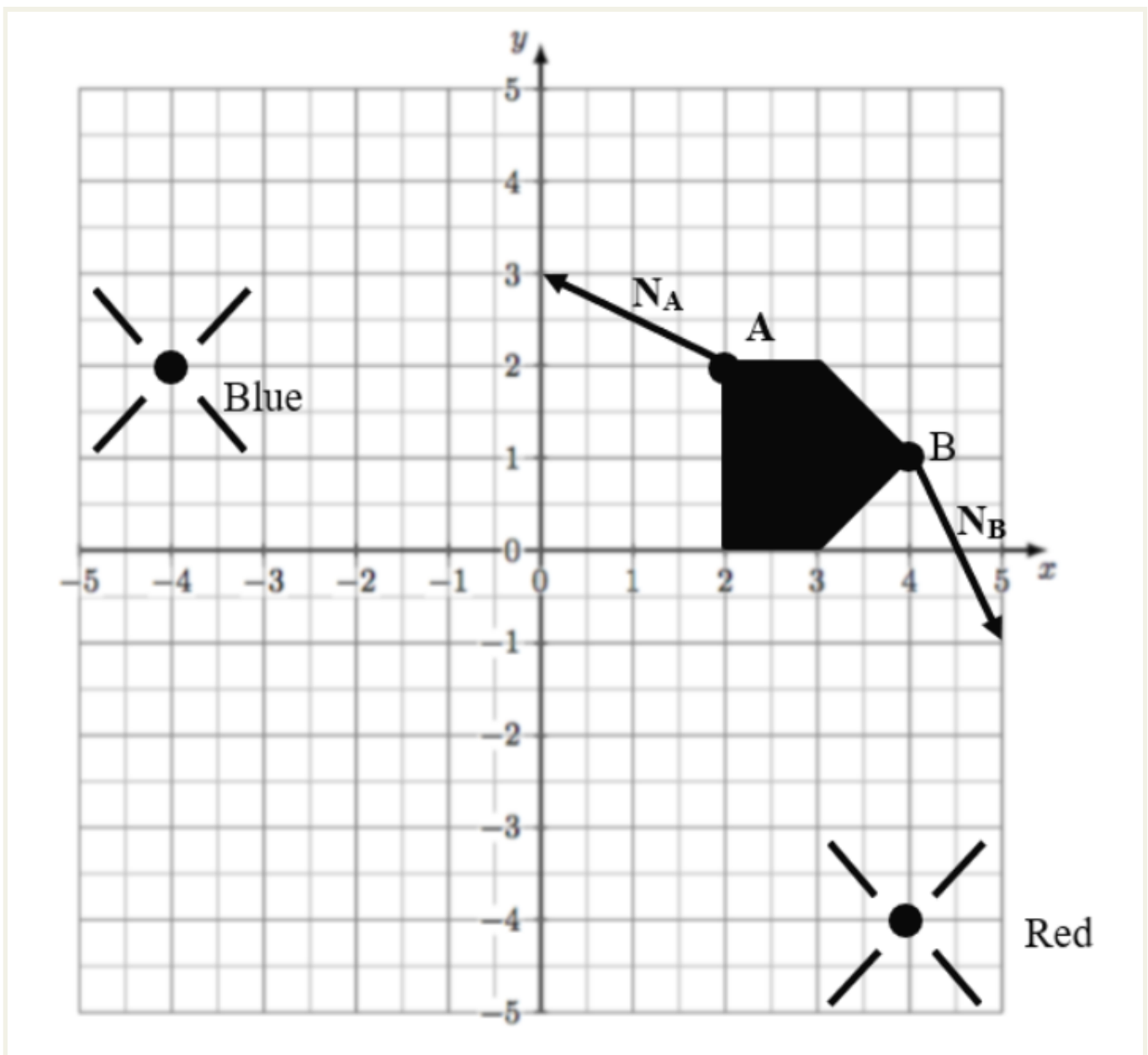
☒ (-1, -2, 0)

4 $\Rightarrow$ 0

The figure below contains two light sources and two points in 2D. The red-light source emits diffuse light = $(1.0, 0.2, 0.2)$ and its spotlight effect = 1. The blue-light source emits diffuse light = $(0.2, 0.1, 1.0)$, spot cutoff = 45, and spot exponent = 2.0. The normal vectors are shown in the figure. Assume the constant attenuation = 0, linear attenuation = 1.0, quadratic attenuation = 0, the material ambient = $(0, 0, 0)$, the material emission = $(0.1, 0.1, 0.1)$, the diffuse components of the material = $(0.5, 1.0, 0.5)$, and the specular components of the material = $(0, 0, 0)$ for the two points. Using the Phong reflection model, answer the following questions:

The diffuse intensity for the point A for the blue-light source = *

(1 Point)



○ $\frac{1}{2\sqrt{10}}$

☐ $\frac{3}{\sqrt{5}}$

☐ $\frac{2}{\sqrt{5}}$

☐ $\frac{3}{\sqrt{10}}$

27

Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glRotatef(-90.0, 0, 0, 1.0);  
glPushMatrix();  
glScalef (-1.0,0.5, -1.0);  
glTranslatef(-1.0f, -2.0f, -3.0f);  
glRotatef(-90.0, 0, 0, 1.0);  
glPopMatrix();  
glRotatef(90.0, 0, 0, 1.0);  
glTranslatef(1.0f, 2.0f, 3.0f); *
```

(1 Point)

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -1 & 0 & 1 \\ -1 & 0 & 0 & 2 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 1 \\ -1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

☐ A

☐ B

☒ C

☐ D

Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);
```

```
glLoadIdentity();
```

```
glScalef(-1.0, 2.0, -1.0);
```

```
glRotatef(90.0, 0, 0, 1.0); *
```

(1 Point)

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -2 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

☐ A

☐ B

☐ C

☒ D

This 3D homogeneous point: (4,3,2,1) corresponds to what Cartesian point: *

(1 Point)

☐ (4,3,2,1)

☐ (3,2,1)

☐ (3/4, 1/2, 1/4)

☒ (4,3,2)

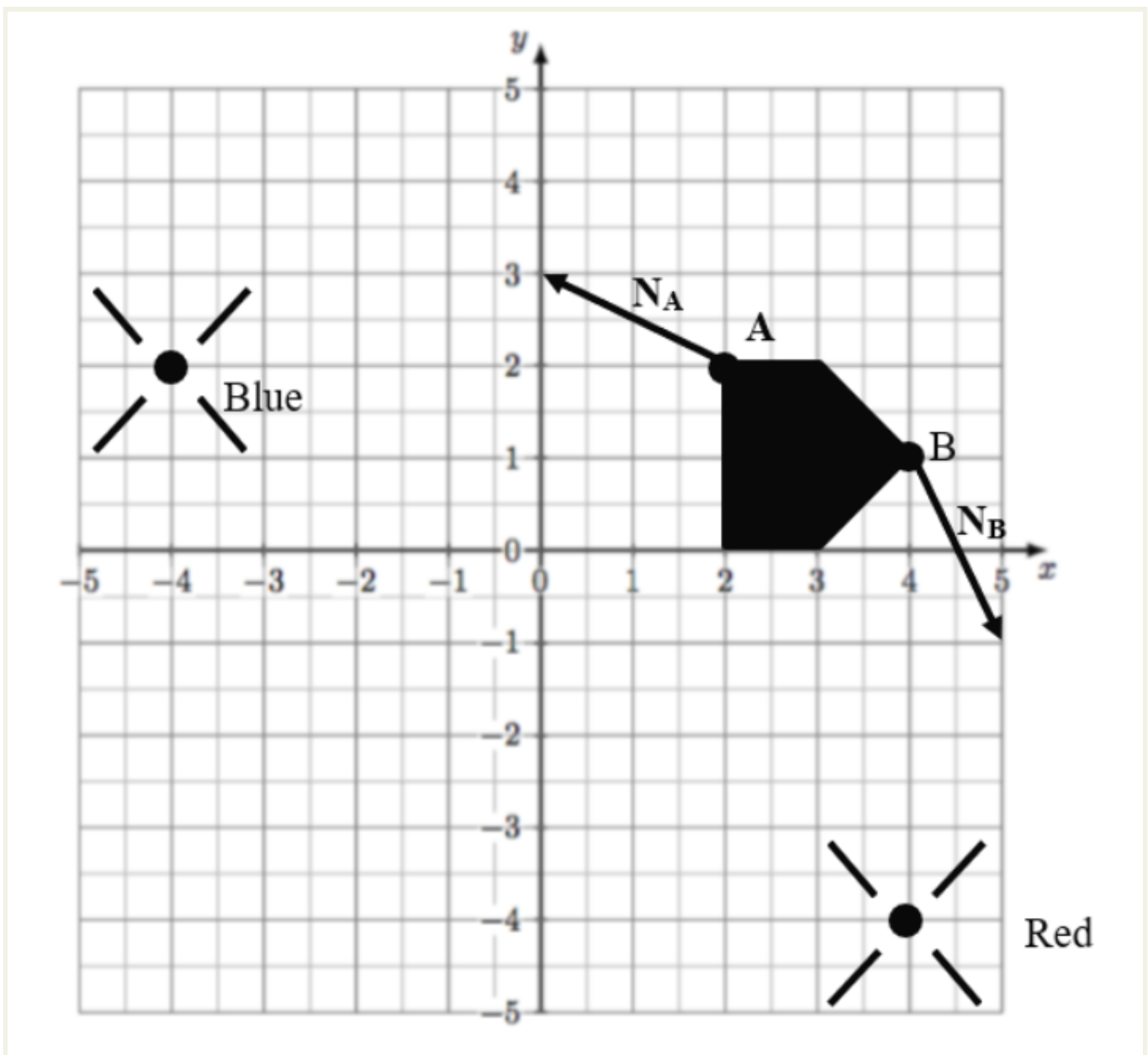
Does OpenGL provide modelling primitives such as a cube, cone, pyramid? *
(1 Point)

- ☐ Yes, but only with glEnable(GL_PRIMITIVES) called before the glBegin statement
- ☐ In the official specification, but also enhanced versions provided by the GLUT library
- ☒ Not in the official specification, but such functionality is provided by the GLUT library.
- ☐ Yes

The figure below contains two light sources and two points in 2D. The red-light source emits diffuse light = $(1.0, 0.2, 0.2)$ and its spotlight effect=1. The blue-light source emits diffuse light = $(0.2, 0.1, 1.0)$, spot cutoff = 45, and spot exponent=2.0. The normal vectors are shown in the figure. Assume the constant attenuation=0, linear attenuation =1.0, quadratic attenuation=0, the material ambient = $(0, 0, 0)$, the material emission= $(0.1, 0.1, 0.1)$, the diffuse components of the material = $(0.5, 1.0, 0.5)$, and the specular components of the material = $(0, 0, 0)$ for the two points. Using the Phong reflection model, answer the following questions:

The attenuation factor for the point A for the blue-light source = *

(1 Point)



○ 6

☐ $\frac{1}{2\sqrt{41}}$

☐ $\frac{1}{2\sqrt{10}}$

32

What components make up a light source in OpenGL? *
(1 Point)

- ☐ Specular and Ambient
- ☐ Diffuse and Ambient
- ☐ Diffuse, Opaque, Ambient
- ☒ Diffuse, Specular, and Ambient

33

Does OpenGL use column-major order or row-major order when representing a transformation matrix? *
(1 Point)

- ☐ row-major
- ☒ column-major

34

What does glMaterialf(...) do? *

(1 Point)

- ☐ Turns material properties on
- ☐ Switches to material mode to add visual effects
- ☐ Gets the current material properties of a drawn object
- ☒ Defines the material properties of a drawn object for visual effects

35

What is the difference between glColor3d and glColor3f? *

(1 Point)

- ☐ glColor3d only sets RGB, while glColor3f sets R, G, B and A
- ☐ glColor3d allows 3d color operations, while glColor3f only allows 8-bit
- ☒ glColor3d takes double arguments, while glColor3f takes float arguments
- ☐ glColor3d is in real space, glColor3f is in integer space

36

Rotating the 2D Cartesian point (0,-1) by 90 degree gives what: *

(1 Point)

- ☒ (1,0)
- ☐ (-1,0)
- ☐ (0,1)
- ☐ (0,-1)

37

The process of removal of hidden surfaces is termed as *

(1 Point)

- ☐ clipping
- ☐ copying
- ☒ culling
- ☐ shorting

38

The OpenGL function used to set up a parallel view volume is *

(1 Point)

- ☐ glutCreateWindow
- ☐ glRectf
- ☒ glOrtho
- ☐ glViewport

39

If you had 1,000,000 vertices and wanted to change all the vertex positions by multiplying each x-coordinate by the same vertex's y-coordinate, the fastest place to do it is: *

(1 Point)

- ☐ with a c++ function
- ☒ in the vertex shader
- ☐ in the fragment shader

40

If lighting is disabled, which function specifies the color of the vertex? *
(1 Point)

- ☐ glClearColor()
- ☐ glDisplayfunc()
- ☒ glColor()
- ☐ None of These

41

The Modelview matrix after executing the following OpenGL code segment is:
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glScalef (-2.0,1.5, -3.0) *
(1 Point)

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & -2 \\ 1 & 0 & 0 & 1.5 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & 1.5 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1.5 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 & -2 & 0 \\ 0 & 1.5 & 0 & 0 \\ -3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- ☐ A
- ☒ B
- ☐ C
- ☐ D

The dot product of any two perpendicular vectors equals one *

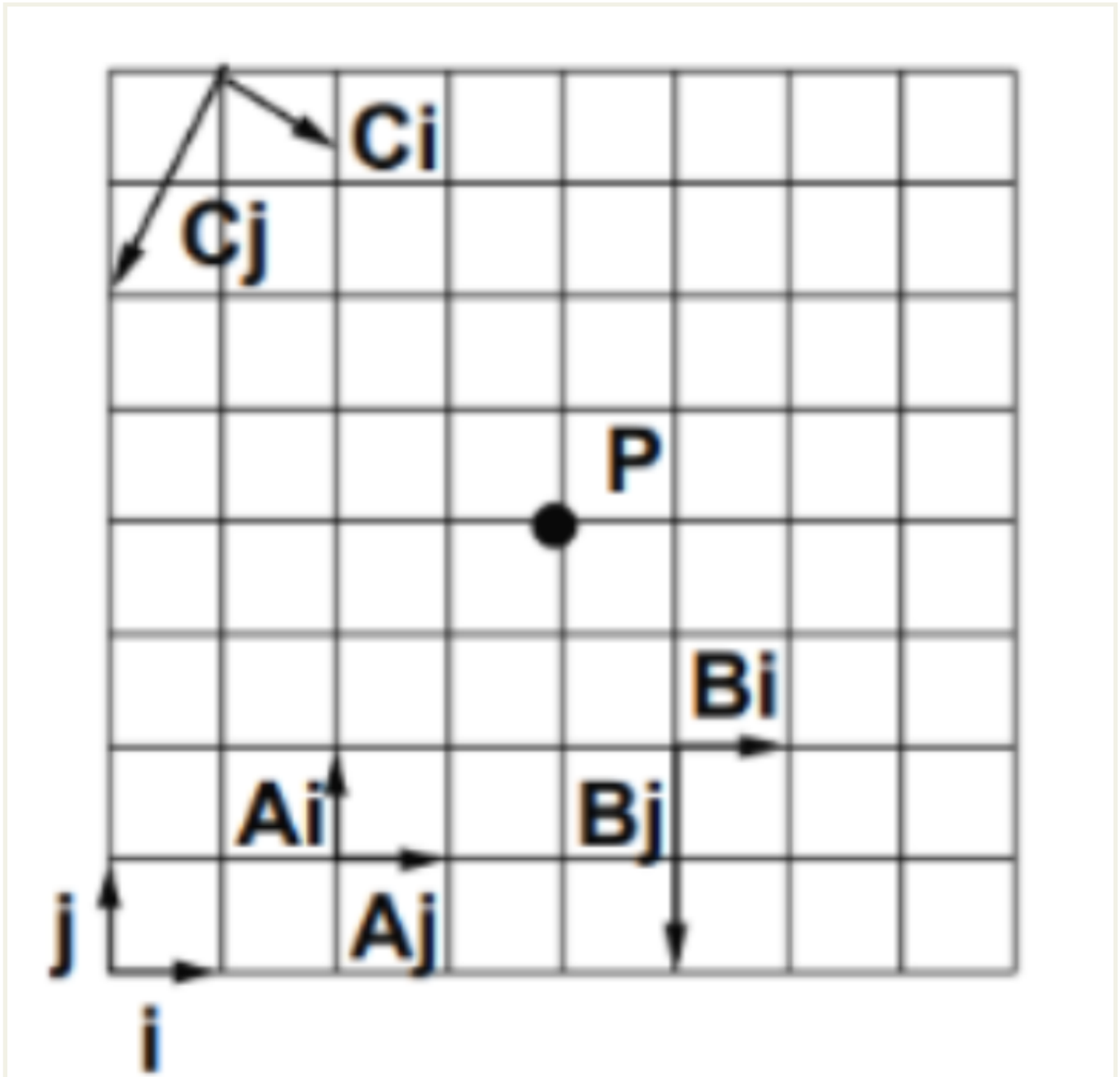
(1 Point)

☐ True

☒ False

The figure contains a point coordinate P that can be expressed as $P = 4*i + 4*j$, where i and j are basis vectors of unit length along the x and y axes, respectively. Using the figure below to answer the following questions.

..... is the coordinates of point P with respect to coordinate frames B^*
(1 Point)



☐ (-1,-1)

☐ (1,2)

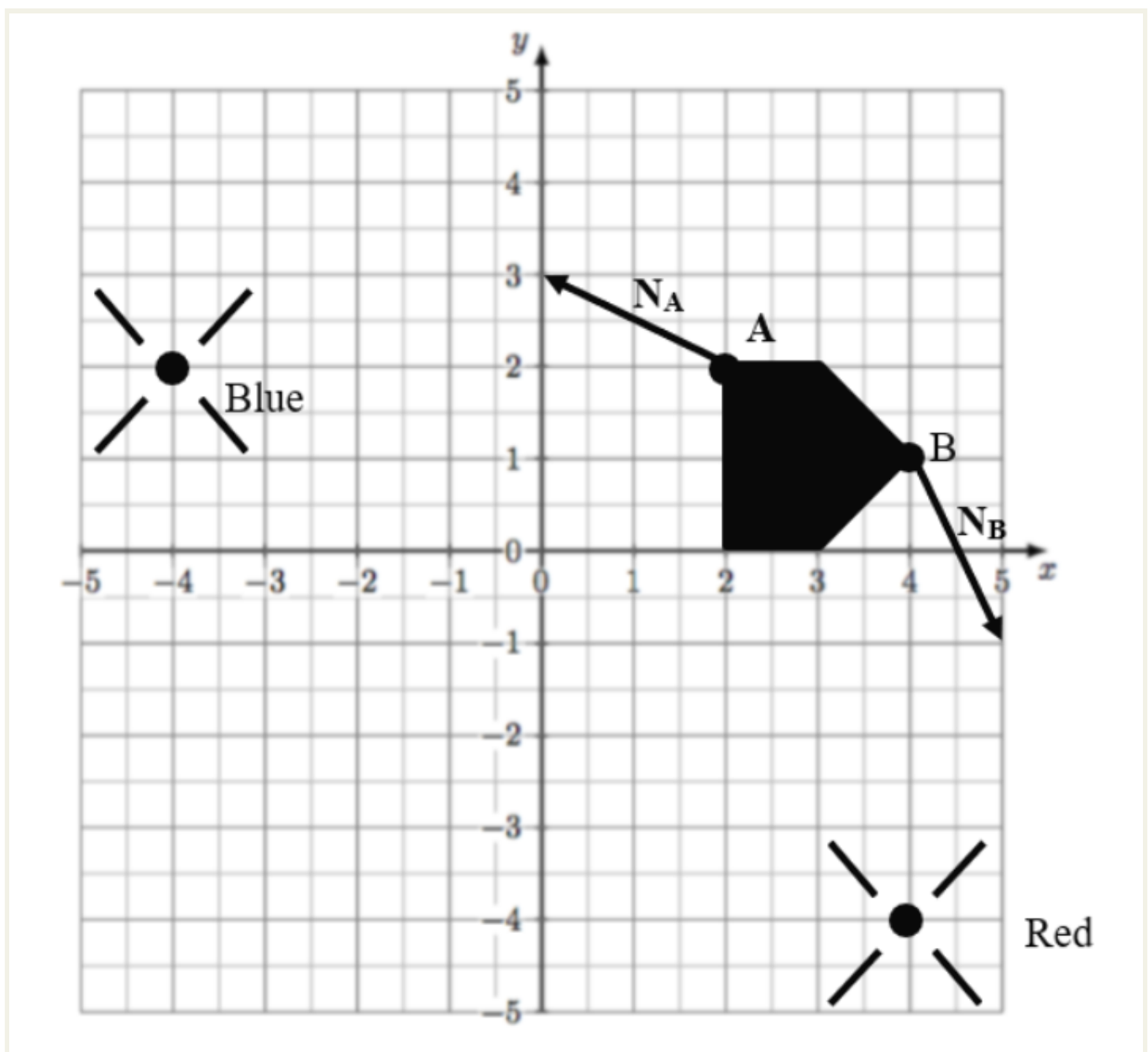
☐ (-2,-2)

☐ (2,1)

The figure below contains two light sources and two points in 2D. The red-light source emits diffuse light = (1.0, 0.2, 0.2) and its spotlight effect=1. The blue-light source emits diffuse light = (0.2, 0.1, 1.0), spot cutoff = 45, and spot exponent=2.0. The normal vectors are shown in the figure. Assume the constant attenuation=0, linear attenuation =1.0, quadratic attenuation=0, the material ambient = (0, 0, 0), the material emission= (0.1,0.1,0.1), the diffuse components of the material = (0.5,1.0,0.5), and the specular components of the material = (0, 0, 0) for the two points. Using the Phong reflection model, answer the following questions:

The normalized normal vector $N_B = *$

(1 Point)



$\bigcirc \left(\frac{1}{\sqrt{3}}, \frac{-2}{\sqrt{3}} \right)$

☐ $\left(\frac{1}{\sqrt{5}}, \frac{-2}{\sqrt{5}} \right)$

☐ $\left(\frac{2}{\sqrt{3}}, \frac{-1}{\sqrt{3}} \right)$

☐ $\left(\frac{2}{\sqrt{5}}, \frac{-1}{\sqrt{5}} \right)$

45

If the scaling factor value is less than one, then *
(1 Point)

- ☒ It reduces the size of object
- ☐ It increases the size of object
- ☐ It does not change the shape of an object
- ☐ None

46

Rotate (z, 90) * Rotate (z, 45) = Rotate (z, 45) * Rotate (z, 90) *
(1 Point)

- ☒ True
- ☐ False

47

Positive value for the rotation angle Θ defines. *
(1 Point)

- ☐ Clockwise rotation about the pivot point
- ☒ Counterclockwise rotation about the pivot point

48

..... light component comes from one direction and once it hits a surface, it is scattered. *

(1 Point)

- ☒ Diffuse
- ☐ Specular
- ☐ Ambient
- ☐ Spot

49

What is the purpose of GL_REPEAT? *

(1 Point)

- ☐ To repeat the last command
- ☐ To enable double buffering
- ☐ To restart the application
- ☒ To have a texture repeat in a direction (horizontal or vertical)
- ☐ To have a command repeat until stopped

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);
```

```
glLoadIdentity();
```

```
glRotatef(90.0, 0, 0, 1.0); *
```

(1 Point)

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

☐ A

☐ B

☐ C

☐ D

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_LINE_LOOP);
```

```
    glVertex2fv(A);
```

```
    glVertex2fv(B);
```

```
    glVertex2fv(C);
```

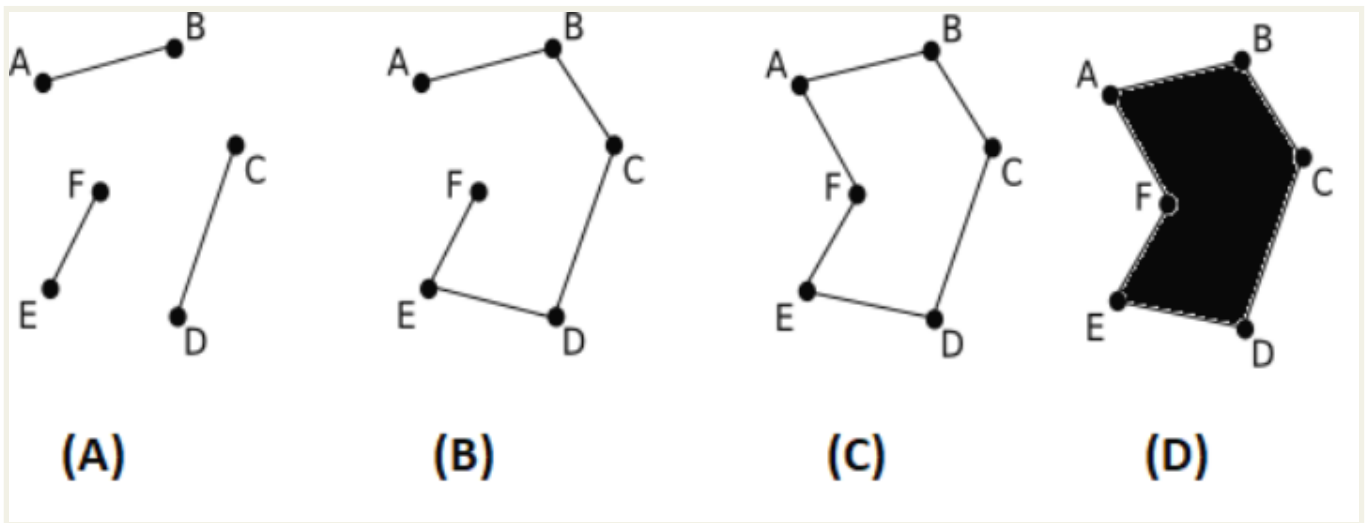
```
    glVertex2fv(D);
```

```
    glVertex2fv(E);
```

```
    glVertex2fv(F);
```

```
glEnd(); *
```

(1 Point)



☐ A

☐ B

☒ C

☐ D

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_LINES);
```

```
    glVertex2fv(A);
```

```
    glVertex2fv(B);
```

```
    glVertex2fv(C);
```

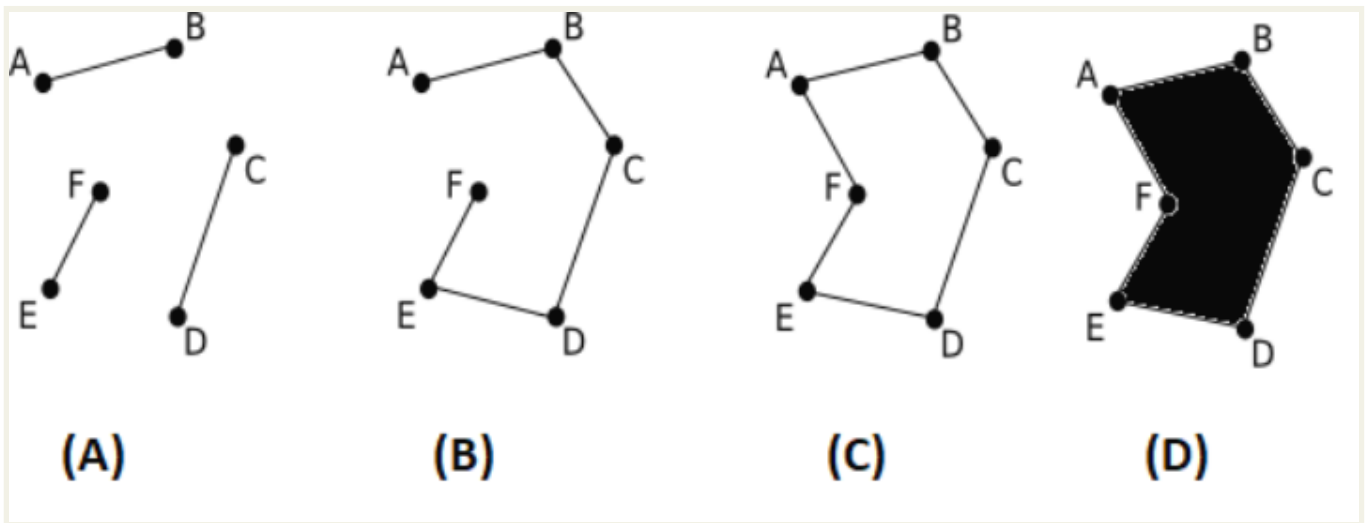
```
    glVertex2fv(D);
```

```
    glVertex2fv(E);
```

```
    glVertex2fv(F);
```

```
glEnd(); *
```

(1 Point)



☒ A

☐ B

☐ C

☐ D

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_LINE_STRIP);
```

```
    glVertex2fv(A);
```

```
    glVertex2fv(B);
```

```
    glVertex2fv(C);
```

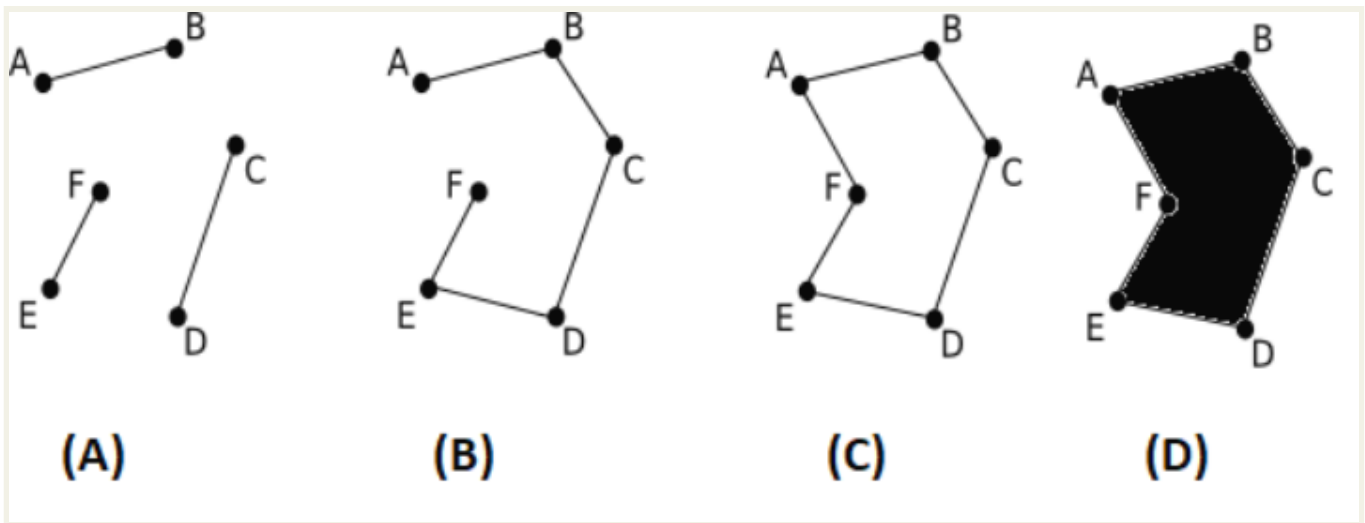
```
    glVertex2fv(D);
```

```
    glVertex2fv(E);
```

```
    glVertex2fv(F);
```

```
glEnd(); *
```

(1 Point)



☐ A

☒ B

☐ C

☐ D

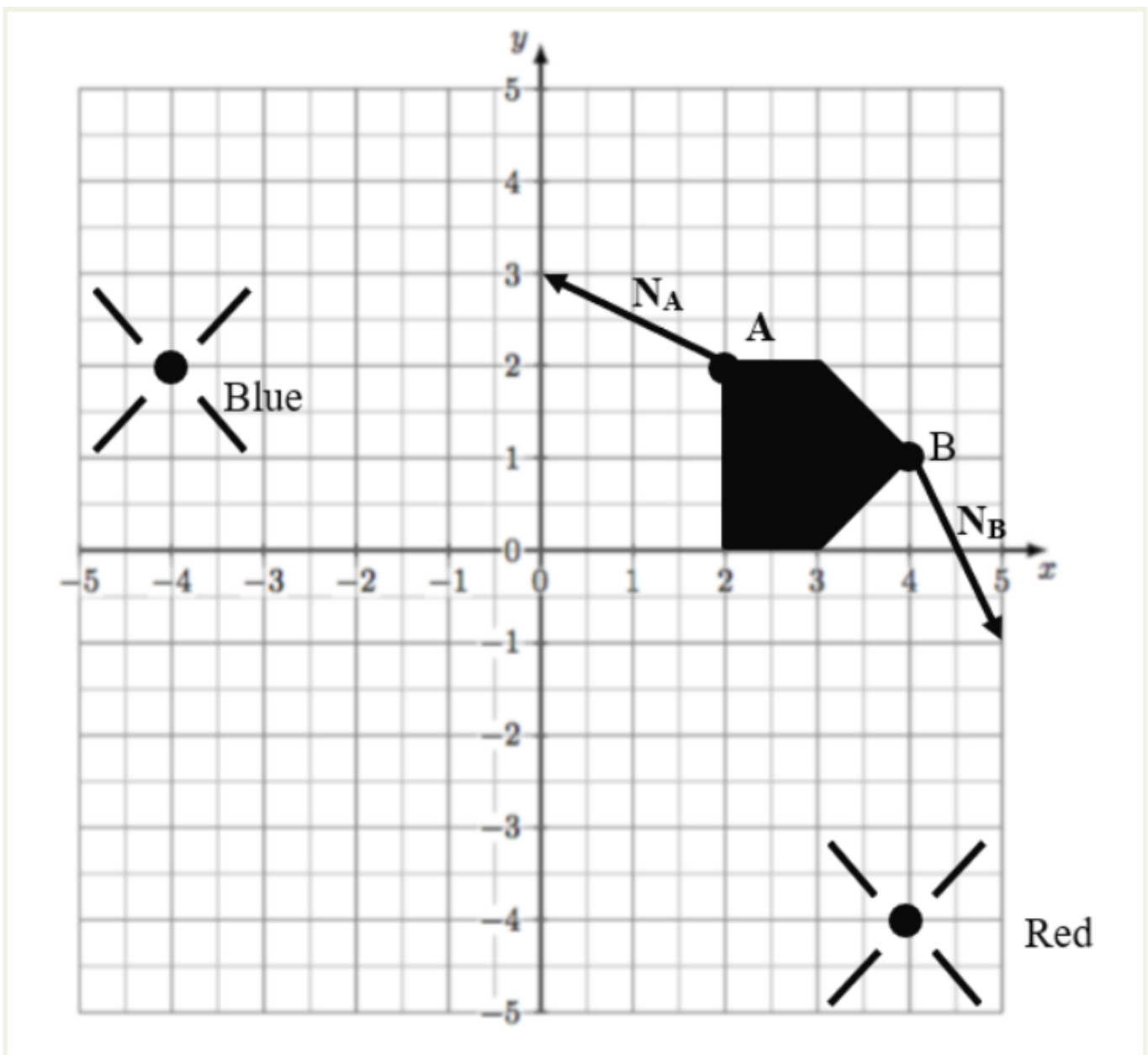
What type of vertex shader variable has the same value for each vertex? *
(1 Point)

- ☒ uniform variable
- ☐ varying variable
- ☐ lighting variable
- ☐ attribute variable

The figure below contains two light sources and two points in 2D. The red-light source emits diffuse light = $(1.0, 0.2, 0.2)$ and its spotlight effect = 1. The blue-light source emits diffuse light = $(0.2, 0.1, 1.0)$, spot cutoff = 45, and spot exponent = 2.0. The normal vectors are shown in the figure. Assume the constant attenuation = 0, linear attenuation = 1.0, quadratic attenuation = 0, the material ambient = $(0, 0, 0)$, the material emission = $(0.1, 0.1, 0.1)$, the diffuse components of the material = $(0.5, 1.0, 0.5)$, and the specular components of the material = $(0, 0, 0)$ for the two points. Using the Phong reflection model, answer the following questions:

The attenuation factor for the point A for the red-light source = *

(1 Point)



$\bigcirc \frac{1}{\sqrt{10}}$

☐ $\frac{1}{\sqrt{40}}$

☐ 1

☐ $\frac{1}{2\sqrt{10}}$

56

What is the normalized surface normal vector to the plane $x+2y+2z+4=0$? *

(1 Point)

☐ (1,2,2)

☐ (1/3,2/3,2/3,4/3)

☐ (1/3,2/3,2/3,-4/3)

☐ (1/3,2/3,2/3)

57

..... light comes from one direction and tends to bounce off the surface in a preferred direction. *

(1 Point)

☐ Diffuse

☐ Specular

☐ Ambient

☐ Spot

The OpenGL command `glClearColor(0.0f, 1.0f, 0.0f, 1.0f)` clear a window with blue color *

(1 Point)

☐ True

☒ False

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